

JONEIL CAOILE

joneilcaoile@gmail.com | (424) 421-9173 | Torrance, CA 90502 | [linkedin.com/in/joneilcaoile](https://www.linkedin.com/in/joneilcaoile) | joneilcaoile.github.io

PROFESSIONAL SUMMARY

M.S. Biomedical Engineering candidate (Stevens, 3.9 GPA) with hands-on experience authoring V&V test protocols, applying signal-based validation, and designing simulated-use studies. Background in physiological signal analysis, bench-level prototyping, and Python-based simulation applied to medical device V&V. Bridges clinical perspective and engineering rigor in V&V protocol design.

TECHNICAL SKILLS

Verification & Validation: Test protocol authoring, acceptance criteria, design verification, simulated-use testing, statistical validation methods, traceability, requirements analysis, IQ/OQ/PQ frameworks (coursework).

Risk & Quality Tools: ISO 14971 risk management (coursework), FMEA (coursework), root cause analysis, design controls, CAPA (coursework), nonconformance investigation, statistical sampling.

Technical: Python (NumPy, SciPy, Matplotlib), MATLAB, R, SQL (SQLite, PostgreSQL), Excel, Git, SolidWorks; EMG signal analysis, motion capture (OptiTrack), physiological data acquisition, biomechanics, cardiopulmonary physiology.

EDUCATION

Stevens Institute of Technology, *M.S. Biomedical Engineering* Expected May 2026 | GPA 3.9
Concentration: Medical Devices & Clinical Development | Provost Master's Scholarship
Relevant Coursework: Machine Learning, Clinical Research and Design, Biofeedback, Physiology, BMI, Biostatistics

California State University, Long Beach, *B.S. Kinesiology: Exercise Science* Dec 2020 | Cum Laude

VERIFICATION & VALIDATION PROJECTS

Chest Drain Air Leak Monitor – Inline Retrofit Cartridge (Advisor: Dr. Jinho Kim) Jan 2026 – May 2026 (Expected)

- Designed an inline retrofit cartridge with an integrated pressure tap (MPRLS + FS3000, dual I2C) that adds digital air leak classification to existing chest drains; thresholds per Cerfolio drainology.
- Defining sensor accuracy acceptance criteria; resolved I2C bus contention with 4.7 kΩ series resistors.
- Designed a SolidWorks adapter (3D printed via Bambu) to interface the FS3000 flow sensor with the test rig for closed-loop bench testing.

Parkinson's Tremor-Suppression System (Product & Prototyping Lead) Sep 2025 – Dec 2025

- Built a Python (NumPy, SciPy) simulation pipeline (4-8 Hz Butterworth bandpass, RMS envelope, proportional closed-loop decoding) to verify EMG-based tremor detection before hardware implementation.
- Achieved ~45% reduction in simulated tremor amplitude (Pearson $r = 0.994$, RMSE ≈ 0.12); defined pass/fail criteria for detection latency and suppression accuracy.

Clinician-Facing Reliability Tool: Fall-Risk Assessment (Advisor: Dr. Antonia Zaferiou) Jan 2025 – May 2025

- Developed a validation tool that cleaned and merged a 77-patient clinical survey dataset (Qualtrics exports) into SQLite to calculate inter-rater reliability metrics (Krippendorff's Alpha, percent agreement, uncertainty rates).
- Built a clinician-facing decision-support dashboard, iterated outputs with faculty stakeholders, and designed the SQLite schema for migration to PostgreSQL.

EXPERIENCE

Graduate Researcher Jan 2025 – Present
Stevens Institute of Technology

- Design test protocols and instrumentation for a retrofit chest drain sensor module capstone in Dr. Jinho Kim's lab (Jan 2026–Present).
- Previously: Dr. Antonia Zaferiou's Musculoskeletal Control and Dynamics Lab (Jan-May 2025); see Reliability Tool project above.

Physical Therapy Aide 2016 – 2023 | 400+ Hours
Various Physical Therapy Clinics (Orthopedic, Pediatric, Geriatric)

- Assisted therapists with patient sessions, device setup, and treatment workflow across three clinical settings. Built first-hand understanding of end-user needs for medical device design and testing.

CERTIFICATIONS

Microsoft / LinkedIn Career Essentials (Apr 2026): Data Analysis, Cybersecurity (95%), Business Analysis (95%), Project Management; Atlassian Agile Project Management (94%); Aha! Product Management Certified.